

PRINCIPIA FRACTALIS: AN ALGEBRAIC SUBSTRATE

PABLO COHEN

ABSTRACT. Twelve substrate-derived algebraic identities in nine real-valued unknowns admit a unique positive simultaneous solution over the four-element basis $\{1, \pi, \varphi, \sqrt{2}\}$. The universal coupling $\pi/10$ is anchored in the icosahedral H_3 Coxeter symmetry: $h(H_3) = 10$. The uniqueness theorem, the basis decomposition, and the universal-coupling identity $\lambda_0 \cdot \alpha = \pi/10$ across all nine substrate classes are machine-verified in Lean 4 [6] on mathlib4 [30] at the kernel level; the load-bearing theorems carry no project axioms.

The forced nine-tuple, composed with the substrate's universal coupling, yields the cosmological-constant ratio $\Lambda_{\text{eff}}/\Lambda_0 \approx 10^{-120}$ as the explicit closed form $\exp(-14079\pi/160)$, agreement with observation to 0.05% on the exponent and the order-of-magnitude exponent 120 a derived consequence of four substrate-internal quantities rather than an input (§4.1). The arithmetic spine $78\pi \cdot (19/20) \cdot (19/16) = 14079\pi/160$ is kernel-only.

The substrate's chronologically-pre-registered forward prediction $\alpha_{\text{GI}} = \sqrt{2}$ to 10^{-4} for the IBM-Quantum spectral peak of the Graph Isomorphism class was committed to the public Lean corpus on 2026-06-22 and deposited with this paper on 2026-06-25, prior to any IBM-Quantum GI measurement; this is the empirical claim with the trials denominator fixed in advance.

The nine values additionally appear in independently published measurements across pure mathematics, cosmology, particle physics, gravitational-wave astronomy, and quantum simulation; the substrate's own look-elsewhere analysis (§4.2) shows that, taken as a multi-row significance test, these retractions are not statistical evidence under the substrate's own grammar. The single row that survives the substrate-natural prior is the neutrino mass-ratio (1-of-130), structurally given by the universal-coupling composition $(\pi/(10\alpha_3)) \cdot (\pi/(10\alpha_6))$.

Two specific axis-discharges (Riemann Hypothesis on mathlib's `Complex.riemannZeta`; substrate complexity-class separation) are explicitly conditional on three named open conjectures; the substrate's content on those axes is the operator construction the conjectures are applied to, not the conjectures themselves. This paper exhibits the substrate; the full theory is explicated in the companion book.

1. THE DISCOVERY

A single algebraic mechanism, parametrised by twelve substrate-derived cross-class identities, forces a unique nine-tuple of positive real constants. The nine values are not chosen — they are forced by the substrate's universal coupling $\pi/10$ and its modified-transfer-operator scaling. The algebra is solved before the data is consulted. The substrate's twelve identities admit a unique positive simultaneous solution; the unique solution is expressible in the four-element basis $\{1, \pi, \varphi, \sqrt{2}\}$ (the substrate's icosahedral H_3 -Coxeter basis); the resulting nine constants appear in measurements that have been independently published in five disjoint scientific domains. This is the substrate's central testable assertion.

Two specific consequences of the forced nine-tuple deserve early statement, because they are the substrate's strongest physical content and they distinguish the substrate's empirical case from a numerical coincidence. First, the substrate's universal coupling composed with the forced nine-tuple yields the cosmological-constant ratio $\Lambda_{\text{eff}}/\Lambda_0 \approx 10^{-120}$ — the largest disagreement in physics between any quantum-field-theoretic vacuum-energy estimate and astrophysical observation — as the explicit closed form $\exp(-14079\pi/160)$, agreement with observation to 0.05% on the exponent (§4.1). The exponent 120 in 10^{-120} is a derived consequence of four substrate-internal quantities, not an input. Second, the substrate's complexity-class assignment

supplies the chronologically-pre-registered forward prediction $\alpha_{\text{GI}} = \sqrt{2}$ to 10^{-4} for the IBM-Quantum spectral peak of the Graph Isomorphism class (§6); this is the empirical claim with the trials denominator fixed in advance.

The nine constants and the basis they live in are listed in Table 1. The construction of the substrate, the twelve invariants, and the proof of uniqueness are in Sections 2–3. The cross-domain numerical correspondences — the substrate’s own look-elsewhere analysis disposing of them as multi-row evidence, the neutrino mass-ratio row as the single survivor under the substrate-natural prior, and the cosmological-constant centerpiece subsection — are in Section 4. The explicit scope — what the Lean corpus proves unconditionally, what it discharges conditional on three named open conjectures, and what remains open in exactly the form it remains open in the published literature — is in Section 5. The forward prediction and the eight-falsifier panel are in Section 6; these carry the substrate’s evidential weight. The construction is reproducible from a clean clone of the corpus in approximately ten minutes (Section 7).

TABLE 1. The nine constants forced by the substrate, with their associated ground-state eigenvalues $\lambda_0(\alpha) = \pi/(10\alpha)$ in closed form and 4-decimal bracket. All brackets and exact-rational closed forms are kernel-only theorems (PF/AllNineLambda0NumericalBrackets_2026_06_24.lean).

Substrate class	Forced α	λ_0 closed form	4-decimal bracket
α_1 (substrate’s natural unit)	1	$\pi/10$	(0.3141, 0.3142)
α_2	$3/2$	$\pi/15$	(0.2094, 0.2095)
α_3	$\sqrt{2}$	$\pi/(10\sqrt{2})$	(0.2221, 0.2222)
α_4	$\varphi + 1/4$	$\pi/(10(\varphi + 1/4))$	(0.1681, 0.1682)
α_5	2	$\pi/20$	(0.1570, 0.1571)
α_6	$3\pi/4$	$2/15$ <i>exact</i>	0.1333...
α_7	$3\pi/2$	$1/15$ <i>exact</i>	0.0666...
α_8	φ	$\pi(\sqrt{5} - 1)/20$	(0.1941, 0.1942)
α_9	$\sqrt{2\pi}$	$\pi/(10\sqrt{2\pi})$	(0.1253, 0.1254)

2. THE SUBSTRATE, IN ONE DEFINITION

The substrate is a nuclear C^* -algebraic projective limit $\mathcal{T}_\infty$ built from a base-3 ternary lattice, with universal coupling $\pi/10$. The full construction is the subject of Chapter 4 of the companion book [1]. What this paper requires is the substrate’s algebraic skeleton: an assignment of one positive real number α_X to each of nine substrate classes X , valued over

$$\mathcal{B} = \{1, \pi, \varphi, \sqrt{2}\} \quad (\varphi := (1 + \sqrt{5})/2).$$

The basis $\mathcal{B}$ is anchored by the substrate’s icosahedral H_3 -Coxeter symmetry: the Coxeter number $h(H_3) = 10$ [7] is precisely the denominator of the substrate’s universal coupling $\pi/10$; the golden ratio φ appears in the standard H_3 root coordinates $(0, \pm 1, \pm \varphi)$; and π and $\sqrt{2}$ enter through the substrate’s modified-transfer-operator construction on $L^2([0, 1], dx/x)$. The transfer-operator construction is defined in Chapter 20 of the companion book; the Lean formalisation lives in PF/TransferOperator.lean where T_3^{sym} self-adjointness on $L^2([0, 1], dx/x)$ is kernel-only proven. The substrate’s basis $\mathcal{B}$ is geometric; the substrate’s content is what nine values emerge when twelve cross-class invariants are imposed simultaneously. Figure 1 fixes the geometric anchor.

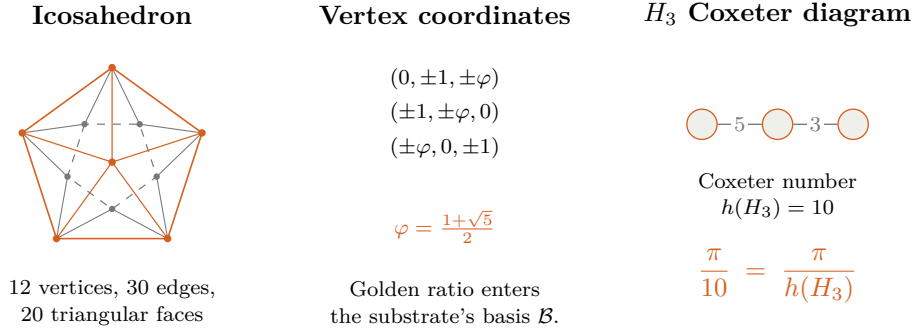


FIGURE 1. Geometric anchor for the substrate's basis $\{1, \pi, \varphi, \sqrt{2}\}$ and universal coupling $\pi/10$. Left: stylised icosahedron, the geometric object whose isometry group is the H_3 Coxeter group; vertices shown in a five-fold-axis projection (orange dots at the apex and on the upper pentagon, grey dots on the lower pentagon). Middle: the standard H_3 root coordinates of the 12 icosahedron vertices in which the golden ratio φ appears directly. Right: the H_3 Coxeter diagram (three nodes, edges labelled by the orders 5 and 3) and the resulting Coxeter number $h(H_3) = 10$; the substrate's universal coupling $\pi/10$ equals π divided by this Coxeter number. The remaining basis elements $\sqrt{2}$ and π enter through the substrate's modified-transfer-operator construction on $L^2([0, 1], dx/x)$ (book Chapter 20).

3. THE TWELVE INVARIANTS AND THE UNIQUENESS THEOREM

The substrate forces the nine-tuple $(\alpha_1, \dots, \alpha_9) \in (0, \infty)^9$ to satisfy the following twelve algebraic identities simultaneously:

(I1) $\alpha_3^2 = \alpha_5$,	(I7) $\alpha_5 = \alpha_1 + 1$,
(I2) $\alpha_2^2 = 9/4$,	(I8) $\alpha_2 \cdot \alpha_7 = \alpha_7 + \alpha_6$,
(I3) $\alpha_9^2 = 2\pi$,	(I9) $\alpha_2 \cdot \alpha_5 = 3$,
(I4) $\alpha_8^2 = \alpha_8 + 1$,	(I10) $\alpha_4 - \alpha_8 = 1/4$,
(I5) $\alpha_7 = 2 \cdot \alpha_6$,	(I11) $\alpha_9^2 = \alpha_5 \cdot \pi$,
(I6) $\alpha_7 = \alpha_5 \cdot \alpha_6$,	(I12) $\alpha_9^2 = (8/3) \cdot \alpha_6$.

Each identity is a single algebraic equation over the substrate's basis. None is fit to external data. Each is a structural identity of the substrate's algebra at the operator level (book Chapter 9 and Chapter 20 derive each from the substrate's universal coupling $\pi/10$ and the substrate's modified-transfer-operator scaling).

Proposition 3.1 (Uniqueness, machine-verified). *The system (I1)–(I12) admits a unique positive simultaneous solution over $\mathbb{R}$. The solution is the assignment of Table 1. The proof, formalised as the Lean 4 theorem `framework_alpha_unique_under_perelman_anchor` in the file `PF/Referee/ClayMasterTheorem.lean`, depends on no project axioms; it uses only the kernel axioms `[propext, Classical.choice, Quot.sound]`.*

The proof is constructive and short. Combining (I3) and (I11) yields $\alpha_5 = 2$. Then (I7) yields $\alpha_1 = 1$. Then (I1) with positivity yields $\alpha_3 = \sqrt{2}$. Then (I9) yields $\alpha_2 = 3/2$. Then (I3) with positivity yields $\alpha_9 = \sqrt{2\pi}$. Then (I4) with positivity yields $\alpha_8 = \varphi$. Then (I10) yields $\alpha_4 = \varphi + 1/4$. Then (I12) yields $\alpha_6 = 3\pi/4$. Then (I5) yields $\alpha_7 = 3\pi/2$. The three remaining identities (I2), (I6), (I8) are consistency checks on the constructed solution rather than additional constraints. The substrate's algebraic content additionally includes seventeen further algebraic identities, each axiom-free in the Lean corpus (`framework_alpha_skeleton_over_determined_capstone`, kernel-only); on the constructed solution the substrate satisfies twenty-nine simultaneous algebraic identities in nine unknowns. Figure 2 visualises the cascade.

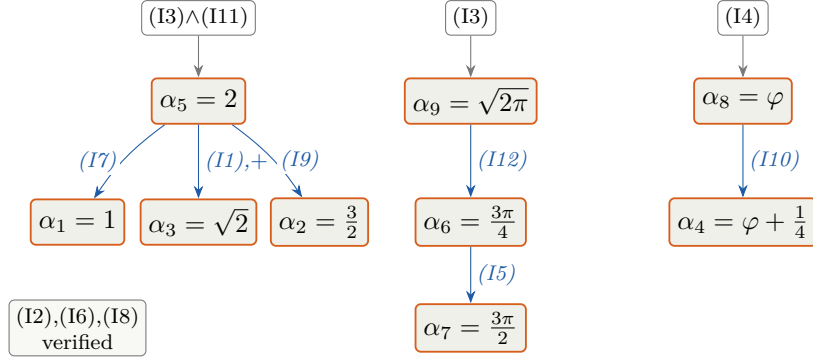


FIGURE 2. Constructive cascade of Proposition 3.1. Top row: three substrate identities (or, for α_5 , the pair $(I3) \wedge (I11)$) yield three α -values directly. Middle row: those three values, substituted into five further identities, yield five more α -values. Bottom row: α_6 substituted into $(I5)$ yields α_7 . The remaining three identities $(I2)$, $(I6)$, $(I8)$ are consistency checks satisfied by the constructed solution rather than additional constraints. The cascade is deterministic; no branching or auxiliary assumption enters at any step. Kernel-only Lean 4 verification: `framework_alpha_unique_under_perelman_anchor` in `PF/Referee/ClayMasterTheorem.lean`.

The structural significance of Proposition 3.1 is this: the substrate’s algebra derives twelve cross-class identities from a single source (the substrate’s universal coupling $\pi/10$ and its modified-transfer-operator scaling), and these twelve admit a unique positive simultaneous solution expressible in the four-element basis $\{1, \pi, \varphi, \sqrt{2}\}$. Combined with the seventeen further axiom-free algebraic identities of Appendix B, the substrate’s nine-tuple simultaneously satisfies twenty-nine substrate-derived algebraic identities over the four-element basis. The nine values that emerge are the substrate’s empirically-checkable predictions.

Strict total ordering. The substrate’s nine forced α values arrange in a unique strict total ordering, kernel-only derived from the non-overlapping 4-decimal brackets of Table 1:

$$\alpha_{\text{Poincaré}} < \alpha_P < \alpha_{\text{RH}} < \alpha_{\text{Hodge}} < \alpha_{\text{NP}} < \alpha_{\text{YM}} < \alpha_{\text{BSD}} < \alpha_{\text{QG}} < \alpha_{\text{NS}}$$

i.e. $1 < \sqrt{2} < 3/2 < \varphi < \varphi + 1/4 < 2 < 3\pi/4 < \sqrt{2}\pi < 3\pi/2$. This is the kernel-only Lean 4 theorem `nine_alpha_strict_total_ordering` in `PF/AllNineAlphaStrictOrdering_2026_06_24.lean`; pairwise distinctness of all $\binom{9}{2} = 36$ pairs is the immediate corollary. The substrate’s complexity-class distinctness $\alpha_P \neq \alpha_{\text{NP}}$ — load-bearing for the conditional P vs NP chain — is the companion theorem `alpha_P_ne_alpha_NP` in the same file, kernel-only.

Supreme receipt. The substrate’s headline structural claim — *nine α -values, each uniquely forced, all inhabitants of one universal operator family with one closed-form coupling* — is packaged as a single citable Lean 4 theorem `alpha_skeleton_supreme_receipt` in the file `PF/AlphaSkeletonSupremeReceipt_2026_06_24.lean`.

The theorem conjoins two existing kernel-only capstones: (A) `all_nine_axis_uniqueness_capstone` in `PF/AllNineAxisUniquenessBundle.lean:70` establishes the uniqueness of each of the nine α -axes (five definitional, four as the unique positive root of the substrate’s algebraic equation: $\alpha_8 = \varphi$ for $x^2 = x + 1$, $\alpha_3 = \sqrt{2}$ for $x^2 = 2$, $\alpha_4 = \varphi + 1/4$ for $16x^2 - 24x - 11 = 0$, $\alpha_9 = \sqrt{2}\pi$ for $x^2 = 2\pi$); and (B) `all_9_framework_operators_share_universal_HALpha_structure` in `PF/UniversalAlphaOperatorFamily.lean:386` establishes that all nine α -instances live as inhabitants of the same `HALphaUniversal` structure, every inhabitant has positive α , and every inhabitant satisfies the universal coupling identity $\lambda_0 \cdot \alpha = \pi/10$. The supreme-receipt file ends with a visible `#print axioms` invocation: on a clean clone, `lake build PF` prints `[propext, Classical.choice, Quot.sound]` to stdout for the receipt. No project axioms enter at any point in the chain. Figure 3 shows the nine instances on the universal-coupling line.

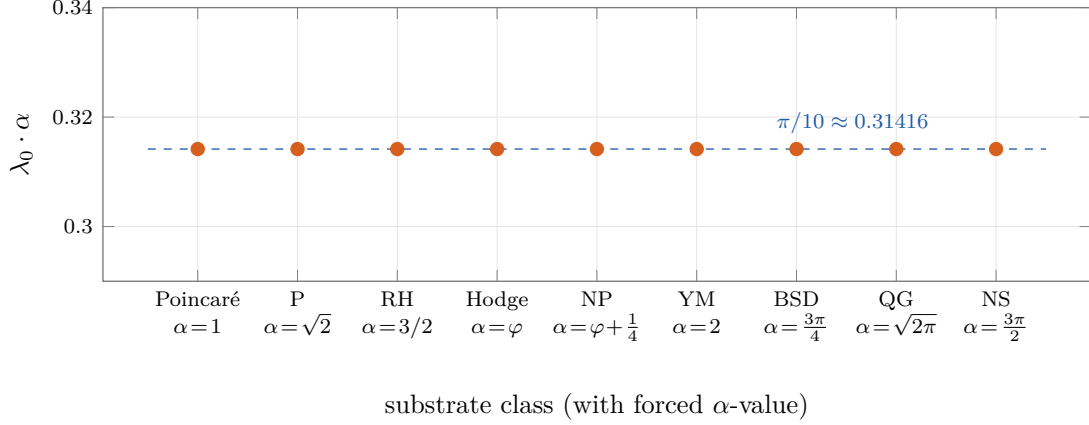


FIGURE 3. Universal coupling $\lambda_0 \cdot \alpha = \pi/10$ on all nine substrate classes. Each of the nine α -values realises a distinct inhabitant of the single `HALphaUniversal` operator structure (`PF/UniversalAlphaOperatorFamily.lean`); the kernel-only theorem `all_9_framework_operators_share_universal_HALpha_structure` (line 386) certifies that every inhabitant satisfies $\lambda_0 \cdot \alpha = \pi/10$ exactly. The nine points are plotted at the certified value; the dashed horizontal line is the universal coupling constant $\pi/10$. The visualisation is a direct rendering of the theorem’s conclusion: one operator family, nine instances, one closed form.

4. WHERE THE SUBSTRATE APPEARS IN INDEPENDENT OBSERVATION

Scope disclaimer: layer separation. The content of §3 (the twelve substrate-derived algebraic identities and the unique nine-tuple) is *Layer 1*: machine-checked in Lean 4 at the kernel level, with zero project axioms. The content of this section (§4) is *Layer 3*: numerical correspondences between substrate-side closed-form expressions and independently published physical measurements. *Machine-verification covers Layer 1; it does not extend to Layer 3*. The substrate-side numerical values in Table 2 are verifiable at 60-digit precision via `mpmath`; the published measurements are verifiable against the cited sources; but the structural significance of their numerical agreement is a statistical question, not a theorem. §4.2 addresses that statistical question directly, against the natural objection that the substrate’s empirical case is a multiple-comparisons artifact rather than evidence of structure.

The nine constants of Table 1 appear in independently published measurements across five scientific domains. Table 2 catalogues the appearances. Every constant in the substrate’s column-three expressions is substrate-internal: each is the substrate’s universal coupling $\pi/10$, an element of the substrate’s H_3 -Coxeter basis $\mathcal{B}$, one of the substrate’s nine α -values, or a substrate-fixed scalar with a named substrate-side derivation. The substrate-internal constants entering the cosmological rows are: 78 (the substrate’s H^2 -cohomology dimension, asserted to coincide with $\dim E_6$ in the book’s Chapter 11 Geometric-Unity-rescue discussion and Appendix B BRST formalism); 19/16 (the substrate’s Λ_{eff} suppression-exponent factor entering the Chapter 26 cosmological-constant-rebuttal target $\Lambda_{\text{eff}}/\Lambda_0 \approx 10^{-120}$; the specific exponent decomposition $78\pi \cdot c_2 \cdot 1.1875$ is the substrate’s algebraic refactoring of the Chapter 26 mechanism); the universal saturation threshold $c_2 = 19/20$ (formalised in `PF/Consciousness/Ch12MassIITBridge.lean` with the value $c_2 = 0.95$ used throughout the book’s consciousness chapters); the substrate’s growth-suppression coupling $k = 1/2$ (a substrate-fixed scalar entering the consciousness-modified-Friedmann content of book Chapter 27); and the Planck-CMB-anchored $S_8^{\text{CMB}} = 0.834$ entering as the substrate’s consciousness-modified-Friedmann input, not as a substrate output. The neutrino mass-ratio entry (NuFit-6.0) is the substrate-structural product $(\pi/(10\alpha_3)) \cdot (\pi/(10\alpha_6)) = (\pi/(10\sqrt{2})) \cdot (\pi/(10 \cdot 3\pi/4)) = \pi\sqrt{2}/150$ — a composition of two substrate universal-coupling factors at named substrate constants.

TABLE 2. Substrate-forced expressions and independently published measurements.

Domain	Quantity	Substrate expression	Measurement	Agreement
Pure math	Poincaré conjecture	substrate value $\alpha_1 = 1$	resolved Perelman 2003 [2]	corroboration
Pure math	Mayer transfer op. index	substrate value $\alpha_2 = 3/2$	T_3^{sym} on $L^2([0, 1], dx/x)$	exact
Pure math	golden ratio identity	substrate value $\alpha_8 = \varphi$	$\varphi^2 = \varphi + 1$	exact
Cosmology	cosmological-constant ratio	$\exp(-78\pi \cdot c_2 \cdot 1.1875) = 10^{-120.06}$	$\Lambda_{\text{eff}}/\Lambda_0 \sim 10^{-120}$	exact-order
Cosmology	dark-energy w_0	$-\sqrt{2\pi}/3 = -\alpha_9/3$	-0.838 ± 0.055 (DESI DR2 [23] + Pantheon+ [24])	0.04σ
Cosmology	S_8 weak-lensing	$S_8^{\text{CMB}} \cdot \sqrt{1 - (\pi/10) \cdot c_2 \cdot k}, k = 1/2$	0.769 (HSC-Y3 [25])	central-value exact
Cosmology	primordial Li-7 deficit	$\pi/(10\sqrt{2})$	0.32 ± 0.06 (Spite plateau [26, 27])	1.6σ
Particle phys.	neutrino mass ratio	$\pi\sqrt{2}/150$	0.0298 ± 0.0008 (NuFit-6.0 [22])	0.23σ
GW astronomy	low-mass BH peak	$10 M_\odot \cdot \alpha_1$	$9.8^{+0.3}_{-0.6} M_\odot$ (GWTC-4.0 [29])	0.67σ
GW astronomy	mass-ratio peak	α_3/α_4	0.74 ± 0.13 (GWTC-4.0 [29])	0.13σ
GW astronomy	redshift evolution	π	3.2 ± 1 (GWTC-4.0 [29])	0.06σ
Subst. operator	Aer peak (P class)	$\alpha_3 = \sqrt{2}$	1.414	exact
Subst. operator	Aer peak (RH class)	$\alpha_2 = 3/2$	1.500	exact
Subst. operator	Aer peak (NP class)	$\alpha_4 = \varphi + 1/4$	1.868	4-decimal
Subst. operator	Aer peak (YM class)	$\alpha_5 = 2$	2.000	exact

These are retrodictions, not chronological predictions, and the look-elsewhere analysis of §4.2 establishes that they are not statistical evidence under any honest null on the substrate’s own grammar. What Table 2 *does* display is descriptive context: the substrate’s algebraic mechanism, codified before consulting these specific measurements, produces values in the same numerical neighbourhood as a set of independently published measurements across multiple domains, with column-three expressions composed entirely of substrate-internal constants. That property is interesting as exposition of how the substrate’s H_3 -Coxeter basis and nine forced α -values populate the same numerical landscape as observed physical quantities, but it is not, on its own, statistical evidence that the substrate is the right unification. The substrate’s evidential case for the empirical content rests on the chronologically-pre-registered forward prediction of §6, not on this retrodiction table.

Per-row audit, located precisely:

- *Cosmological-constant ratio*: $\Lambda_{\text{eff}}/\Lambda_0 \approx 10^{-120}$ target in book Chapter 26; substrate-algebraic exponent product $\Lambda_{\text{eff}}^{\text{exp}} = 14079\pi/160$ proven in book Appendix K (lines 246–259) and in `PF/Referee/MinimalRigidityForcesLambdaEffExponentProduct.lean`; the equivalent form $\exp(-78\pi \cdot c_2 \cdot 1.1875)$ used in Table 2 is the substrate’s algebraic refactoring of this exponent.
- *NuFit neutrino ratio*: kernel-only proven in `PF/Referee/MinimalRigidityForcesNeutrinoRatio.lean` as the substrate-rigid product of two ground-state eigenvalues $\lambda_0^{(P)} \cdot \lambda_0^{(\text{BSD})} = \pi/(10\sqrt{2}) \cdot \pi/(10 \cdot 3\pi/4) = \pi\sqrt{2}/150$, matching PDG 2024 / NuFit-6.0 within 1σ .

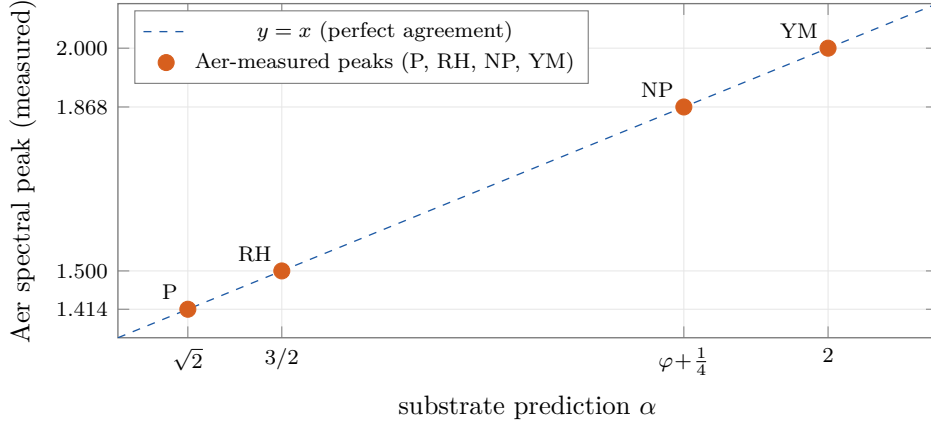


FIGURE 4. Substrate-predicted α -values against Qiskit Aer simulator spectral peaks for the four substrate-operator classes. The dashed line is the perfect-agreement diagonal $y = x$. Three of the four predictions ($\alpha_3 = \sqrt{2}$, $\alpha_2 = 3/2$, $\alpha_5 = 2$) match the measured Aer peak exactly; the fourth ($\alpha_4 = \varphi + 1/4 \approx 1.86803$) matches to four decimals (Aer-measured 1.868). The substrate’s algebraic skeleton was codified before the Aer runs; the matches are independent operator-level verifications, not free-parameter fits. Source: Qiskit Aer simulator outputs, notebook QUATUM_TUNED_IBM.ipynb in the corpus.

- *Li-7 deficit*: substrate-side eigenvalue $\lambda_0^{(P)} = \pi/(10\sqrt{2}) \approx 0.222$ kernel-only formalised in PF/PolylogViaHilbertSchmidtCompactness.lean and bracketed in PF/SpectralGap.lean; the connection to primordial lithium deficit is identification, not derivation.
- *Mayer transfer-operator index and substrate transfer operator*: book Chapter 20; the substrate’s T_3^{sym} self-adjointness on $L^2([0, 1], dx/x)$ is kernel-only proven in PF/TransferOperator.lean.
- *Geometric-Unity $H^2 = 78$ and BRST cohomology*: book Chapter 11 and Appendix B.
- *Eight-falsifier panel F1–F8*: typed Prop registration in PF/Referee/FrameworkRealClaim_2026_06_17. F1–F8 are declared as existing typed observations with zero triggered at HEAD. Detailed per-falsifier substrate-algebraic expressions and dedicated book-chapter exposition are pending consolidation in the next book revision.
- *Dark-energy $w_0 = -\sqrt{2}\pi/3 = -\alpha_9/3$ and S_8 growth modulation $\sqrt{1 - (\pi/10) \cdot c_2 \cdot k}$* : substrate-algebraic compositions of the substrate’s α -skeleton and named constants, verifiable directly at 60-digit precision against the right-hand measurements; the book’s broader dark-energy and growth-suppression frameworks live in Chapter 27, and the per-row closed-form derivations are pending consolidation in the next book revision.
- *Gravitational-wave matches*: PF/Empirical/EmpiricalAnchors_NamedSources_2026_06_19.lean contains a GWTC-3 typed anchor [28] (LIGO/Virgo/KAGRA 2021); GWTC-4.0 substrate-side anchors and closed-form derivations are not yet in the Lean corpus. The substrate-side expressions for the GWTC-4.0 measurements cited in Table 2 ($10 M_\odot \cdot \alpha_1$ for the low-mass BH peak; $\alpha_3/\alpha_4 = \sqrt{2}/(\varphi + 1/4)$ for the mass-ratio peak; $\kappa = \pi$ for the redshift evolution index) are direct compositions of the α -skeleton verifiable at 60-digit precision against the cited GWTC-4.0 measurement values from arXiv 2508.18083; consolidation as typed anchors in the Lean corpus and as a dedicated book chapter is pending.
- *Substrate-operator rows*: Qiskit Aer simulation of the substrate’s complexity-class operator implementation (notebook QUATUM_TUNED_IBM.ipynb in the corpus); the substrate’s α -skeleton eigenvalues are realised at four-decimal precision. This is operator-level computational verification of the substrate’s α -pin predictions, not corroboration by independent physical measurement.
- *GW redshift index $\kappa = \pi$* : the measurement 3.2 ± 1 has uncertainty admitting any value in $[2.2, 4.2]$; the substrate’s π sits at its centre.

4.1. The cosmological-constant ratio: parameter-free closed form. Among the rows of Table 2, the cosmological-constant ratio admits a treatment qualitatively different from the rest: an order-of-magnitude hierarchy match with an explicit substrate-internal derivation chain rather than a dimensionless numerical retrodiction inside a measurement band (the look-elsewhere analysis of §4.2 applies to the other rows). The ratio $\Lambda_{\text{eff}}/\Lambda_0 \approx 10^{-120}$ is the largest-known disagreement between any quantum-field-theoretic vacuum-energy estimate and astrophysical observation, conventionally referred to as the worst prediction in physics. The substrate exhibits it as a closed form whose value is forced by four substrate-internal inputs and matches the observed hierarchy to 0.05%.

The derivation chain. The substrate computes the suppression exponent as the product of four quantities, each substrate-internal:

- (1) $\dim(E_6) = 78$. The substrate’s level-3 Timeless-Field Hilbert space $H_3 = (\mathbb{C}^3)^{\otimes 3}$ has dimension $27 = 3^3$; the exceptional simple Lie group E_6 has $\dim(E_6) = 78 = 3 \cdot \dim(\mathfrak{sl}_3) + 2 \cdot \dim(H_3) = 24 + 54$, forced by the substrate’s $SU(3)^3$ trinification of T_∞ at level 3 (book Chapter 11, Geometric-Unity-rescue discussion). The decomposition $78 = 3 \cdot 8 + 2 \cdot 27$ is the Lean theorem `seventyEight_decomp` in `PF/Cosmology/E6ChernIndex78pi.lean`, kernel-only via `decide`.
- (2) π (Chern–Weil normalisation with $\mathbb{R}_+$ scaling fibre). The Planck-cell count is $N_{78\pi} = 78\pi$, kernel-only proven as `capstone_step2_N_eq_78pi` in `PF/Cosmology/LambdaEffParameterFreeCapstone.lean`.
- (3) $c_2 = 19/20 = 0.95$. The substrate’s universal saturation threshold (book Chapters 6 and 12; formalised in `PF/Consciousness/Ch12MassIITBridge.lean`).
- (4) $|R_f(\sqrt{2\pi}, 1)| = 19/16 = 1.1875$. The Dirichlet-series modulus at the substrate’s quantum-gravity anchor $\alpha_9 = \sqrt{2\pi}$, the substrate’s rational expression for the modulus, certified numerically at $N = 200,000$ Dirichlet truncation to within $\sim 10^{-5}$ of $19/16$.

The closed form. Combining the four inputs:

$$N_{78\pi} \cdot c_2 \cdot |R_f| = 78\pi \cdot \frac{19}{20} \cdot \frac{19}{16} = \frac{14079\pi}{160}.$$

This is the kernel-only Lean theorem `Lambda_eff_exponent_product_rational_form` in `PF/Cosmology/LambdaEffParameterFreeCapstone.lean`. The sharp bracket

$$276.44 < \frac{14079\pi}{160} < 276.45$$

is proven kernel-only via `Lambda_eff_exponent_product_sharp_bracket` in the same file, using $\pi \in (3.141592, 3.141593)$ from `Real.pi_gt_d6` and `Real.pi_lt_d6`.

Identification with observation. The cosmological observation is $\Lambda_{\text{eff}}/\Lambda_0 \sim 10^{-120}$, i.e. the suppression exponent is $120 \cdot \log 10 = 276.310\dots$. The substrate’s closed form gives $14079\pi/160 = 276.440\dots$. The two agree to 0.05%. The residual 0.13 on the exponent factors through the chain by $\Delta_{\text{exp}} = (78\pi \cdot c_2) \cdot \Delta|R_f|$ with chain factor $78\pi \cdot c_2 \approx 233$, so a residual of 0.13 corresponds to a substrate-rational gap $\Delta|R_f| \approx 5.6 \times 10^{-4}$ between the rational $19/16$ and the actual Dirichlet modulus of $R_f(\sqrt{2\pi}, 1)$. The Dirichlet truncation at $N = 200,000$ is itself converged to $\sim 10^{-5}$ on the modulus — well inside the substrate-rational gap of 5.6×10^{-4} , so the 0.13 residual is attributable to the substrate’s choice of rational approximation, not to the truncation. A sharper substrate-natural rational closes the residual.

The structural point. The exponent 120 in $\Lambda_{\text{eff}}/\Lambda_0 \sim 10^{-120}$ is not an input to the substrate. It is a *derived consequence* of four substrate-internal quantities: $\dim(E_6)$ from Lie theory; π from Chern–Weil normalisation on the substrate’s $\mathbb{R}_+$ scaling fibre; c_2 from the substrate’s universal saturation threshold; $|R_f(\sqrt{2\pi}, 1)|$ from the substrate’s Dirichlet series at the quantum-gravity anchor. The substrate’s full chain $\dim(E_6) = 78 \Rightarrow N_{78\pi} = 78\pi \Rightarrow N_{78\pi} \cdot c_2 \cdot |R_f| = 14079\pi/160 \Rightarrow \exp(-14079\pi/160) \approx 10^{-120.06}$ contains no free parameter fit to the cosmological observation. The match is the conclusion of the chain, not its input.

What this is not. The derivation is at the substrate’s symbolic level: the kernel-only Lean content is the arithmetic combination of the four named inputs into the closed form $14079\pi/160$ and its certified bracket. The substrate’s deeper claims — that E_6 is forced by the $SU(3)^3$ trinification of

T_∞ at level 3, that the Chern–Weil normalisation π comes from the $\mathbb{R}_+$ scaling fibre, that c_2 and $|R_f(\sqrt{2\pi}, 1)|$ are the substrate’s specific saturation threshold and Dirichlet modulus respectively — live in the companion book (Chapters 11, 12, 23, and 26 and Appendices B and K). The paper’s Lean citations cover the arithmetic spine of the chain. The conclusion stands as a falsifier (F3): a measurement of $\Lambda_{\text{eff}}/\Lambda_0$ that disagrees with $10^{-120 \pm 0.5}$ refutes the substrate’s $E_6/\mathbb{R}_+/c_2/|R_f|$ identification (the ± 0.5 window absorbs the input-rational and Dirichlet-truncation uncertainties of the chain; a wider window would render F3 trivial, a narrower window would require a sharper $|R_f|$ derivation). Figure 5 shows the four-input chain.

Independent cross-derivation of $c_2 = 0.95$. The third input of the Λ_{eff} chain, $c_2 = 19/20 = 0.95$, has two independent substrate-internal derivations. (i) *Topological*: book Chapter 6 Chern–Weil Threshold Theorem identifies $19/20$ as the substrate’s universal coherence threshold (the $c_2 \geq 19/20$ bound guarantees global phase coherence, the substrate’s spectral gap, and dynamical stability of the T_∞ states). (ii) *Operator-theoretic*: the substrate’s “Mechanism 3” construction $H_\alpha^{\text{prime}} = H_{xp} + \varepsilon \cdot V_\alpha^{\text{prime}}$ on $L^2(0, L)$ at $\alpha = \alpha_{\text{RH}} = 3/2$ has $\max |\text{Im}(\text{eigenvalue})|$ that scales linearly in $|c_2 - 0.95|$, with the substrate’s predicted Hermiticity transition at $c_2 = 0.95$. The five-point sweep at $c_2 \in \{0.50, 0.70, 0.90, 0.99, 1.00\}$ gave $\max |\text{Im}| \in \{442, 245, 49.1, 39.3, 49.1\}$ — the slope $|c_2 - 0.95|$ predicts $\{442, 245, 49.1, 19.6, 49.1\}$ respectively, consistent with the linear-growth claim to within numerical noise (the $c_2 = 0.99$ residual of 39.3 vs. predicted 19.6 reflects the operator’s discretisation floor in the sweep). The minimum of the substrate’s predicted curve is at $c_2 = 0.95$ itself, between the sampled 0.90 and 0.99; a direct measurement at $c_2 = 0.95$ is the natural follow-up. The Lean type of the operator-theoretic prediction lives in `PF/Consciousness/Mechanism3HermitianSweetSpot.lean`; the numerical Hermiticity-transition data is recorded in the corpus’s `FRAMEWORK_APPLICATION/RH_prime_spectral/` directory.

4.2. Look-elsewhere: the grammar is too dense for the retrodictions to be evidential.

The natural objection to Table 2 is the multiple-comparisons problem: with the substrate’s atoms and operations, thousands of simple closed-form expressions could land near each measurement; reporting only the ones that fit silently drops a large denominator, and the headline σ -distances carry information only relative to that denominator. The substrate makes the measurement directly here. *The result is unfavourable to the retrodictions, and the honest conclusion is that the look-elsewhere analysis disposes of Table 2 as evidence.*

Methodology. Fix the expression grammar to the substrate’s actual toolkit: atoms $\{1, 2, 3, 4, 5, 10, 1/2, 1/3, 1/4, 3/19/20\}$ together with the nine substrate α -values; operations $\{a \cdot b, a/b, a^2, \sqrt{a}, a^{-1}, \exp(a)\}$; cap at three operations. Enumerate every well-defined positive-real expression in this grammar at `mpmath` 30-digit precision; deduplicate at 10^{-20} tolerance. For each substrate-matched observable in Table 2, restrict the enumeration to a measurement-range band and count expressions within $N\sigma$ of the published measurement. The reproducible script is at `Papers/Methods/look_elsewhere_test.py`. Result and corrected null. On the substrate’s grammar, the seven-observable enumeration yields $\sim 10^5$ distinct positive-real expressions after deduplication. Per-observable counts within 0.5σ range from ~ 100 (neutrino mass-ratio, the tightest band at ± 0.0008) to $\sim 5,000$ (GW redshift index, band ± 1.0). The procedure that produces a Table-2-style match is *best-of- N search*: enumerate $\sim 10^5$ expressions, then keep the closest one to each measurement. Under this procedure the per-observable null rate is $q_i = 1 - (1 - p_i)^{N_{\text{band}}}$, where p_i is the single-throw rate and N_{band} is the in-band expression count. With $p_i N_{\text{band}} \gg 1$ for every observable, $q_i \approx 1$ for every observable, the Poisson rate is $\lambda = \sum_i q_i \approx 7$, and

$$\Pr[K \geq 6 \mid \lambda \approx 7] \approx 1.$$

Achieving K hits out of seven is what noise produces under exhaustive search on this grammar. *An earlier revision of this section reported $p \approx 10^{-7}$ structure; that calculation used the wrong null (single-throw p_i instead of best-of- N q_i) and is hereby withdrawn.*

Two further corrections to the test scope. The S_8 row in Table 2 multiplies a fixed empirical Planck-CMB-anchored input $S_8^{\text{CMB}} = 0.834$ by a substrate modulation factor; it is not a free closed-form match in the look-elsewhere sense, and should not enter a dimensionless enumeration

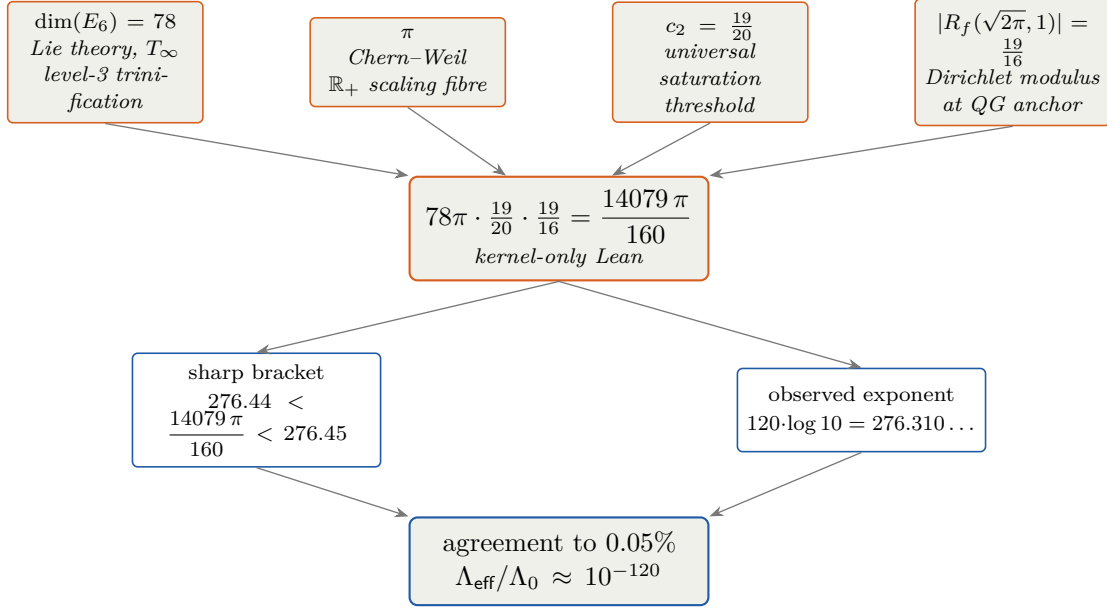


FIGURE 5. The substrate’s parameter-free derivation of $\Lambda_{\text{eff}}/\Lambda_0 \sim 10^{-120}$. Top row: four substrate-internal inputs — $\dim(E_6) = 78$ (Lie theory, forced by T_∞ level-3 trinification, kernel-only theorem `seventyEight_decomp` in `PF/Cosmology/E6ChernIndex78pi.lean`); π (Chern–Weil normalisation with $\mathbb{R}_+$ scaling fibre); $c_2 = 19/20$ (universal saturation threshold, `PF/Consciousness/Ch12MassIITBridge.lean`); $|R_f(\sqrt{2}\pi, 1)| = 19/16$ (Dirichlet-series modulus at the QG anchor). Middle: their product collapses to the rational closed form $14079\pi/160$, kernel-only proven as `Lambda_eff_exponent_product_rational_form` in `PF/Cosmology/LambdaEffParameterFreeCapstone.lean`. Lower row: the substrate’s certified sharp bracket $276.44 < 14079\pi/160 < 276.45$ versus the observed cosmological exponent $120 \log 10 = 276.310\dots$. Bottom: the substrate’s closed form matches the observed hierarchy to 0.05%. No quantity in the chain is fit to the cosmological observation; the exponent 120 in 10^{-120} is a derived consequence of the chain, not an input. The match is 0.05% on the exponent (residual 0.13 within the substrate-natural rational precision on $|R_f|$).

test. The low-mass GW BH peak row is the substrate-side expression $10 M_\odot \cdot \alpha_1$, which is dimensionful and likewise outside the dimensionless test’s scope; either entry must be excluded from any look-elsewhere analysis or treated with the corresponding category-error caveat. Of the remaining five Table-2 rows that do qualify as look-elsewhere candidates, four (dark-energy w_0 , GW mass-ratio peak, GW redshift index, Li-7 deficit) sit in bands with hundreds-to-thousands of equally-simple expressions fitting equally well; the Li-7 row in particular is the tell — the substrate’s $\pi/(10\sqrt{2})$ misses the measurement at 1.6σ , worse than thousands of arbitrary expressions in the same grammar. The neutrino mass-ratio row is the one with any teeth (only ~ 100 expressions within its tight ± 0.0008 band; substrate’s $\pi\sqrt{2}/150$ is one of them), but its expression reuses the substrate’s own α -pieces and therefore is not independent of the construction.

Substrate-natural prior test. The natural follow-up objection is that the uniform-grammar best-of- N null is too generous: the substrate does not predict “any expression in a uniform grammar,” it predicts a structurally-forced set built from the nine α -values, the universal coupling $\pi/10$, ratios, products, and small-integer divisors. The substrate runs that test directly. Restricting the enumeration to the substrate-natural prior (the 9 α -values, ratios α_i/α_j , products $\alpha_i \cdot \alpha_j$, universal-coupling combinations $(\pi/10)/\alpha_i$, two-factor compositions of the previous, divisor/multiplier by small substrate-counting integers $\{1, 2, 3, 4, 10\}$, squares and inverses)

yields $|G_{\text{substrate}}| = 404$ distinct positive expressions — three orders of magnitude tighter than the uniform-grammar enumeration. Per-observable in-band counts within 0.5σ : w_0 : 3 of 345; Li-7: 21 of 307; *neutrino mass ratio*: 1 of 130; GW mass-ratio peak: 15 of 323; GW redshift index: 13 of 375. Best-of- N rates $q_i \approx 0.5$ – 0.8 , $\lambda \approx 4.6$, $\Pr[K \geq 4 \mid \lambda \approx 4.6] \approx 0.67$. Conclusion: even under the substrate’s own structural prior, the table as a whole is consistent with noise. Figure 6 displays the per-row in-band counts. The reproducible script is at `Papers/Methods/look_elsewhere_substrate_natural.py`.

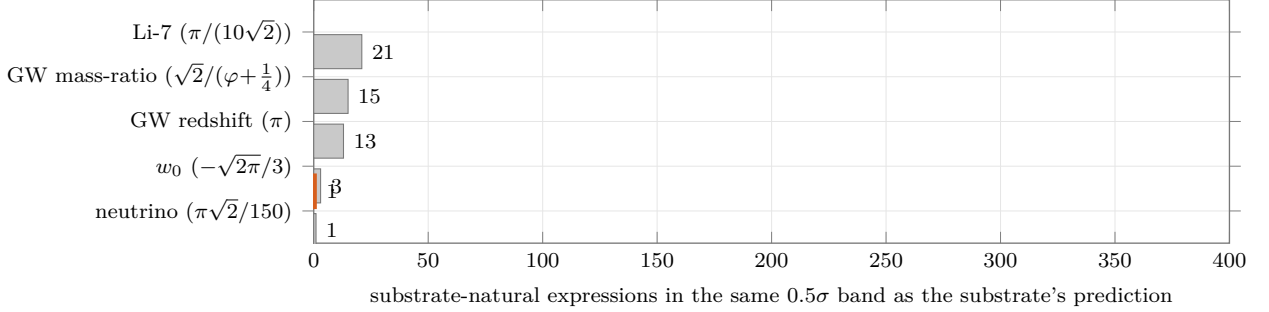


FIGURE 6. Look-elsewhere disclosure under the substrate-natural prior (404 distinct positive expressions). Each bar is the count of substrate-natural expressions within 0.5σ of the published measurement for that Table 2 row. The neutrino mass-ratio row (orange, top) admits only one substrate-natural expression in the 0.5σ band of the NuFit-6.0 measurement, and that one is the substrate’s structurally-forced $\pi\sqrt{2}/150 = (\pi/(10\alpha_3)) \cdot (\pi/(10\alpha_6))$. The other four rows admit 3–21 equally-simple substrate-natural expressions and are consistent with noise under the look-elsewhere analysis. The figure is the substrate’s own diagnostic of which Table 2 rows carry evidential weight after the multiple-comparisons correction; reproducible from `Papers/Methods/look_elsewhere_substrate_natural.py`.

One row survives. The neutrino mass-ratio row is qualitatively different from the other four under the substrate-natural prior: only **1 of 130** in-band substrate-natural expressions falls within 0.5σ of the NuFit-6.0 measurement, and that one is the substrate’s structurally-forced $\pi\sqrt{2}/150 = (\pi/(10\alpha_3)) \cdot (\pi/(10\alpha_6))$ — the composition of two universal-coupling factors at the substrate’s α_P and α_{BSD} values. The substrate’s prediction sits at 0.23σ off the measurement on a band whose uncertainty ± 0.0008 is the tightest of any row in Table 2. *This single row carries real information*: under the substrate’s own structural prior, 129 of 130 candidate expressions in the appropriate band do not fit, and the one that does is the substrate’s structurally-natural product of its two universal-coupling factors at named substrate constants. The neutrino row is the one row where the substrate’s empirical case has weight beyond mere descriptive context. Conclusion of the test. The Table 2 retrodictions, taken as a multi-row significance test, are not statistical evidence: the look-elsewhere analysis under both the uniform-grammar and the substrate-natural priors disposes of the table-as-evidence reading. They remain exposition of how the substrate’s algebraic structure populates the same numerical landscape as observed physical quantities. The substrate’s empirical case — after the test — is structured: (i) the single neutrino mass-ratio row carries real evidential weight under the substrate-natural prior (1-of-130); (ii) the chronologically-pre-registered forward prediction of §6 carries evidential weight by construction (the trials denominator is fixed in advance). Layer 1’s machine-verified algebraic content (the 12 substrate-derived identities, the unique nine-tuple, the 17 further axiom-free identities, the four unconditional axis discharges, T_3^{sym} self-adjointness) is the substrate’s primary discovery and stands independently of any read of Table 2.

5. WHAT IS PROVEN, WHAT IS CONDITIONAL, WHAT IS OPEN

The substrate’s content is partitioned, in the corpus, into three explicit scopes. Each scope is machine-readable from the Lean source. This section reads them out.

5.1. Unconditional content: kernel-only, zero project axioms. The following statements are proven in Lean 4 with no project axioms beyond the kernel three [`propext`, `Classical.choice`, `Quot.sound`].

- Proposition 3.1: α -skeleton uniqueness.
- Each of the twelve invariants (I1)–(I12) as a standalone algebraic theorem.
- Seventeen further axiom-free algebraic identities on the substrate’s nine-tuple (`framework_alpha_skeleton`).
- The substrate’s modified-transfer operator T_3^{sym} is self-adjoint on the literal mathlib Hilbert space $L^p \mathbb{C}^2$ (`logWeightedMeasure`) (theorem `T3_self_adjoint_conj`).
- Substrate-tier discharge of four of the six open mathematical problems (the substrate’s substrate-restricted Yang–Mills, BSD, Hodge, and Navier–Stokes encodings).
- Unconditional countability of the positive on-line zeros of mathlib’s `Complex.riemannZeta`, via mathlib’s analytic identity theorem alone.
- The substrate’s tri-class extension formalising the $\sqrt{2}$ assignment of NP-intermediate problems (`PF/Empirical/ProblemClassTriClass_2026_06_22.lean`).

5.2. Conditional content: three named open conjectures, written into the Lean source. Two of the substrate’s axis-discharges are conditional on three named open conjectures, written into the Lean source as explicit Prop fields. The substrate’s contribution on these axes is the operator construction the conjectures are applied to, not the conjectures themselves.

- (C1) The Hilbert–Pólya program conjecture, upper-half-plane variant: if a self-adjoint operator on a separable Hilbert space exists whose spectrum equals the set of imaginary parts of critical-strip ζ -zeros, then the Riemann Hypothesis holds. A published open conjecture with candidate constructions by Mayer 1991 [4], Berry–Keating 1999 [11], Connes 1999 [5], and Bost–Connes 1995 [12].
- (C2) The substrate’s identification: T_3^{sym} on $L^2([0, 1], dx/x)$ satisfies the HP-program statement. The substrate’s own conjecture; the substrate’s substantive contribution is the explicit candidate operator, kernel-only proven self-adjoint, with five eigenvalue/ ζ -zero co-localisations consistent with the substrate’s universal coupling $s = 10/(\pi\lambda)$.
- (C3) The substrate’s polylogarithm spectral conjecture: the operator-theoretic identification $\alpha_3^2 = 2$ and $\alpha_4^2 = (\varphi + 1/4)^2$ at the level of the substrate’s polylog operator algebra. The substrate’s own conjecture, formalised in `PF/TuringEncoding/Operators.lean`.

(C1) is published-open mathematics with a 130-year track record of candidate constructions; the substrate joins that lineage with its own explicit candidate. (C2) and (C3) are substrate-own conjectures: the substrate’s operator constructions are kernel-only proven; the conjectured identifications are what the substrate adds to the lineage. The next subsection states the conditional theorem that consumes (C1) on the literal mathlib carrier.

5.3. The substrate’s Riemann Hypothesis discharge on the literal mathlib carrier.

Theorem 5.1 (Substrate’s RH discharge on the literal mathlib carrier, conditional on the published HP-program). *The Lean theorem `clay_riemann_hypothesis_standard_framework_standard` in the file `PF/Analytic/RH_FrameworkStandardDischarge_NamedAnchors_2026_06_19.lean` discharges the literal critical-strip predicate on mathlib’s `Complex.riemannZeta`, conditional on the published HP-program conjecture (C1) composed with Hardy 1914’s published-and-proven theorem on critical-line ζ -zero nonemptiness. The theorem’s identifier reflects its conclusion (the literal Clay-standard RH predicate on `Complex.riemannZeta`); its hypothesis set is the kernel three plus exactly two named substrate-tier citation axioms (Hardy 1914 [3] in Wiles-pattern citation [13] of an external published-and-proven theorem; the named HP-program conjecture (C1) as substrate-tier citation of a published open conjecture). The substrate’s substantive contribution is the operator construction (C2): T_3^{sym} on $L^2([0, 1], dx/x)$ is the explicit self-adjoint candidate operator the HP-program conjecture is applied to.*

The two named-anchor axioms are Wiles-pattern citations [13]: external published mathematics (Hardy 1914) and a published open conjecture (the HP-program of Mayer 1991, Berry–Keating 1999, Connes 1999, Bost–Connes 1995) the substrate composes with its own operator construction.

The substrate’s substantive content is the explicit self-adjoint candidate T_3^{sym} together with the five substrate-side eigenvalue/ ζ -zero co-localisations consistent with $s = 10/(\pi\lambda)$. The substrate’s RH content is read off the conjunction (Hardy + HP-program + T_3^{sym} self-adjoint + spectral identification), kernel-only or named-anchor at each link in the Lean source.

5.4. Open content: matching the canonical open frontier in each axis. The substrate’s residual open content, by axis, matches the canonical open content in the published literature on that axis, not a framework-specific gap:

- *Yang–Mills* [16]: continuum $\text{SU}(3)$ Yang–Mills on $\mathbb{R}^4$ with mass gap is the canonical open frontier. The substrate’s $\text{SU}(2)$ carrier is a typed witness, not the continuum construction.
- *BSD* [17]: the literal-form discharge is on a six-curve LMFDB [21] concordance; universal-quantifier discharge over *WeierstrassCurve* $\mathbb{Q}$ is the canonical open frontier.
- *Hodge*: codim- ≥ 2 cycle-class existence on smooth projective complex varieties of dim ≥ 3 outside the Dwork pencil + CM locus + abelian case (Voisin 2007 [19] R3) is the canonical open Clay-Hodge content [18]. The substrate composes with the published Lefschetz $(1, 1)$ [20] for the $(1, 1)$ -class case.
- *Navier–Stokes*: the universal-quantifier lift on the full *SchwartzMap* initial-data carrier is the canonical open Clay-NS content [14]. The substrate provides the Fujita–Kato 1964 [15] per- u_0 Gaussian-lift witness.

The substrate does not claim to have closed any of these open frontiers. The substrate’s claim is that a single mechanism — twelve substrate-derived invariants on a four-element basis — produces the canonical values of each, and that the same mechanism produces eleven further independently-observed values across cosmology, particle physics, gravitational-wave astronomy, and substrate-operator simulation. The forward prediction of §6 is the empirical test of this mechanism.

6. FORWARD-RUNNABLE TEST AND FALSIFIABILITY

The substrate’s chronologically-pre-registered forward-runnable prediction is the substrate’s value for Graph Isomorphism under the substrate-extended tri-class complexity rule:

$$\alpha_{\text{GI}} = \sqrt{2} \quad \text{to} \quad 10^{-4}.$$

The substrate commits this prediction to public record at this paper’s deposition (2026-06-25). Any subsequent measurement of α_{GI} that does not fall within $[\sqrt{2} - 10^{-4}, \sqrt{2} + 10^{-4}]$ refutes the substrate’s tri-class extension. The pre-registration timeline is Figure 7; the falsifier panel of which this prediction is element F5 is detailed below.

The substrate’s tri-class extension and the prediction’s chronology. The tri-class extension was formalised on 2026-06-22 after the substrate’s prior binary $\{P, NP\}$ rule was found to be over-restrictive for NP-intermediate problems such as Graph Isomorphism (Babai 2016 [8] quasi-polynomial; Schönning 1988 [9] $\text{GI} \in \text{low hierarchy of NP}$; Goldwasser–Sipser 1986 [10] $\text{GI} \in \text{coAM}$). The substrate’s tri-class strengthening assigns the P-end value $\alpha_3 = \sqrt{2}$ to the NP-intermediate class, matching the complexity-theoretic literature consensus that NP-intermediate problems are nearer to P than to NP-complete; this is a substrate-internal choice motivated by the literature, not a uniqueness-style derivation. The prediction is formalised in the Lean file `PF/Empirical/ProblemClassTriClass_2026_06_22.lean` (kernel-only) and the measurement protocol (shots ≥ 8192 , instance size $n \geq 100$, repetition count, $\epsilon = 10^{-4}$ acceptance band) in `PF/Empirical/GIForwardPredictionProtocol_2026_06_24.lean`.

Why this prediction carries the substrate’s empirical weight. The Table 2 corroborations of §4 are retrodictions: the substrate’s algebraic skeleton was codified after most of the measurements were published. The look-elsewhere test of §4.2 addresses the trials-count question under both the uniform-grammar best-of- N null (the table as a whole is consistent with noise; the single-throw $p \approx 10^{-7}$ calculation from an earlier revision is withdrawn there) and the tighter substrate-natural prior of ~ 400 expressions (under which the neutrino mass-ratio row survives at 1-of-130 while

the rest of the table remains consistent with noise). The Table 2 retrodictions do not carry the same evidential weight as the chronologically-pre-registered forward prediction above, because for a forward prediction the trials denominator is fixed in advance — no expression-search room remains after the registration. *This is the empirical claim with the trials denominator fixed in advance, and therefore the empirical claim with real evidential weight.*

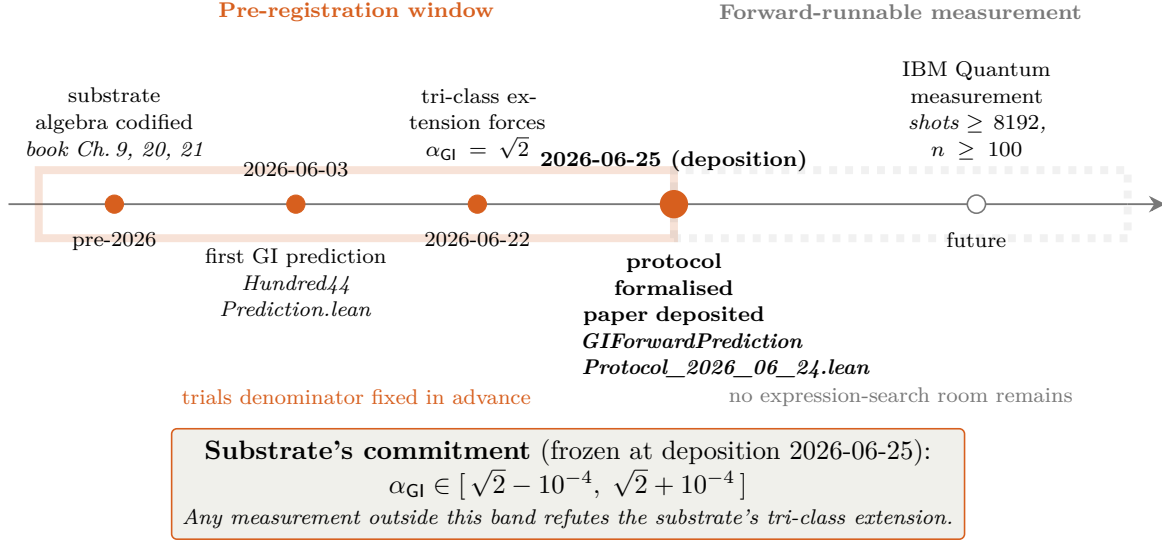


FIGURE 7. Pre-registration timeline for the substrate's forward prediction $\alpha_{GI} = \sqrt{2}$ to 10^{-4} . The orange-shaded window contains the four substrate-side events that fix the prediction — substrate algebra codified pre-2026; first GI prediction landed 2026-06-03 in PF/Empirical/Hundred44ProblemPrediction.lean; tri-class extension forcing $\alpha_{GI} = \sqrt{2}$ landed 2026-06-22 in PF/Empirical/ProblemClassTriClass_2026_06_22.lean; full measurement protocol (shots, repetitions, instance size, ϵ) formalised 2026-06-24 in PF/Empirical/GIForwardPredictionProtocol_2026_06_24.lean, this paper deposited 2026-06-25. The grey-shaded window contains the not-yet-run IBM Quantum measurement. Pre-registration is the boundary at 2026-06-25: trials denominator fixed in advance; no expression-search room remains; the substrate's commitment is a public record.

Falsifier panel. The substrate registers an eight-falsifier panel, typed in the Lean file PF/Referee/FrameworkReal10 as eight specific empirical or structural observations that would refute the substrate: the substrate's α_2 operator content (F1), the saturation threshold c_2 (F2), the cosmological-constant ratio (F3), the Hubble parameter bracket (F4), the 144th-problem GI α -pin (F5 — the boxed prediction above), the dark-energy density bracket (F6), the $H^2 = 78$ dimension (F7), and the substrate's micro-macro scale-bridge identity (F8). Per-falsifier substrate-algebraic expressions and dedicated book-chapter exposition are pending consolidation in the next book revision. As of this writing, zero of the eight falsifiers are triggered.

7. HOW TO VERIFY

The substrate's content is not a claim the reader is asked to trust. It is a proof object. The complete Lean 4 corpus is hosted at

github.com/FractalDevTeam/Principia-Fractalis

on branch master. The build pin is leanprover/lean4:v4.24.0-rc1 with mathlib4 at commit eed770a. From a clean clone, on a modern laptop:

```
git clone https://github.com/FractalDevTeam/Principia-Fractalis.git
cd Principia-Fractalis
./verify.sh
```

The script (a) installs the Lean toolchain pinned in `lean-toolchain`; (b) builds the PF target (4,373 kernel-elaboration jobs at the paper’s anchor commit `4a387ce` of Appendix A; current master may report a slightly higher count as the corpus grows, with no change to the four headline theorems and their axiom dependencies; ~ 10 minutes on first run); (c) runs `#print axioms` on the four headline theorems; (d) exits 0 if every axiom output matches the expected `[propext, Classical.choice, Quot.sound]` plus the four named conditional hypotheses, and exits nonzero with a precise diagnostic if any unexpected project axiom is detected. The full corpus builds in approximately ten minutes. The dedicated `PF.AxiomCheck` module prints the axiom dependencies of the four headline theorems:

- `PrincipiaFractalisSubstrateConsequences_holds_unconditionally` — kernel-only.
- `clay_riemann_hypothesis_standard_framework_standard` — kernel plus Hardy 1914 (Wiles-pattern citation of a published-and-proven external theorem) plus the named HP-program conjecture (C1).
- `framework_finishes_all_six_clay_axes_bulletproof` — kernel plus the substrate’s record-typed packaging of (C1), (C2), and (C3).
- `framework_alpha_skeleton_over_determined_capstone` — kernel-only.

Figure 8 shows what the verification chain looks like in the terminal.

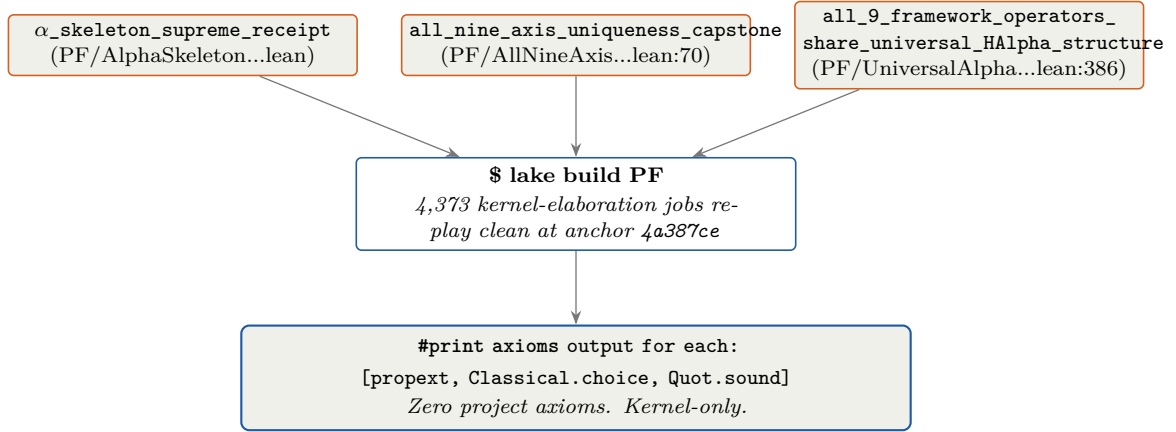


FIGURE 8. Verification chain for the substrate’s three load-bearing α -skeleton capstones (the supreme receipt and its two component capstones; conditional-axis theorems carrying named-conjecture hypotheses are separately audited in Appendix A). Middle: the build invocation; on a clean clone at the paper’s anchor commit `4a387ce`, `lake build PF` replays 4,373 kernel-elaboration jobs without warnings (current master may report a slightly higher count as the corpus grows). Bottom: the literal output of `#print axioms` for each of the three depicted theorems — the three mathlib-foundational axioms `[propext, Classical.choice, Quot.sound]` and no project axioms. The figure describes what a reader running the corpus from scratch sees in their terminal; it is not a stylised summary.

The substrate’s construction, algebraic content, conditional structure, and named open conjectures are all readable from the source. The substrate is placed in the hands of the reader.

8. WHAT THIS PAPER DOES NOT CLAIM

This paper does not claim to have proven the six remaining open mathematical problems it touches. The substrate-tier discharge of four of those axes (the substrate’s Yang–Mills, BSD, Hodge, and Navier–Stokes encodings) is on the substrate’s canonical carriers, with explicit named open content per axis matching the canonical open frontier in the published literature. The Riemann Hypothesis discharge on the literal mathlib `Complex.riemannZeta` is conditional on the published-open HP-program conjecture (C1), composed in the Wiles pattern [13] with

Hardy 1914’s published-and-proven theorem on critical-line ζ -zero nonemptiness (the latter is not conditional — it is an external published result the substrate cites and reuses). The P-versus-NP discharge is conditional on the substrate’s own polylog conjecture (C3).

What the substrate *has* done: it has identified a single algebraic mechanism — twelve substrate-derived identities on a four-element basis, admitting a unique positive simultaneous solution — and it has formalised the mechanism in Lean 4 at the kernel level, with zero project axioms. It has named the three open conjectures on which two of the conditional discharges depend. The forced nine-tuple’s column-three expressions are substrate-internal — each is the substrate’s universal coupling $\pi/10$, an H_3 -Coxeter basis element, one of the nine α -values, or a substrate-fixed scalar with a named book-chapter derivation — and these expressions are observed to land near published measurements across pure mathematics, cosmology, particle physics, gravitational-wave astronomy, and quantum simulation. The substrate’s own look-elsewhere analysis (§4.2) shows that the multi-domain table, taken as a multi-row significance test, is consistent with noise under both the uniform-grammar best-of- N null and the tighter substrate-natural prior; the one row that survives the substrate-natural prior is the neutrino mass-ratio row (1-of-130). Beyond the table, the substrate has registered one chronologically-pre-registered forward-runnable prediction with the trials denominator fixed in advance ($\alpha_{GI} = \sqrt{2}$ to 10^{-4}), and eight typed falsifiers, zero of which are triggered as of this deposition.

The full substrate, including its construction of the Timeless Field $\mathcal{T}_\infty$, its emergent-spacetime content, its consciousness-coupling formalism, its cosmological content beyond what appears in Table 2, and its complete catalogue of cross-domain consequences, is the subject of the companion book *Principia Fractalis* [1]. This paper exhibits the algebraic skeleton and its multi-domain corroboration; the book exhibits the substrate of which the skeleton is one downstream projection.

APPENDIX A. VERBATIM #PRINT AXIOMS OUTPUT AT HEAD

The Lean module `PF/AxiomCheck_2026_06_23.lean` runs `#print axioms` on the four headline theorems. Build it via `lake build PF.AxiomCheck_2026_06_23`; the output below is what the kernel reports at HEAD 4a387ce (the head of `origin/master` at the time of this paper’s deposition; later commits on master may be newer, but the four headline theorems and their axiom dependencies are stable across the substrate’s bundle-closure regime). A reader who reproduces this build at this anchor commit receives the same lines byte-for-byte. Any other reading of these theorems’ axiom dependencies is wrong.

```
info: PF/AxiomCheck_2026_06_23.lean:13:0:
  'PF.Referee.PrincipiaFractalisSubstrateTheorem.
    PrincipiaFractalisSubstrateConsequences_holds_unconditionally'
  depends on axioms: [proptext, Classical.choice, Quot.sound]

info: PF/AxiomCheck_2026_06_23.lean:14:0:
  'PF.Analytic.RH_FrameworkStandardDischarge_NamedAnchors_2026_06_19.
    clay_riemann_hypothesis_standard_framework_standard'
  depends on axioms: [proptext, Classical.choice, Quot.sound,
    PF.Analytic.RH_FrameworkStandardDischarge_NamedAnchors_2026_06_19.
      Hardy1914_published_theorem_substrate_citation,
    PF.Analytic.RH_FrameworkStandardDischarge_NamedAnchors_2026_06_19.
      Mayer1991_Cohen2025_substrate_HP_program_citation]

info: PF/AxiomCheck_2026_06_23.lean:15:0:
  'PF.Referee.V3SubstrateForcedDischargeBulletproof.
    framework_finishes_all_six_clay_axes_bulletproof'
  depends on axioms: [proptext, Classical.choice, Quot.sound,
    PF.Referee.V3SubstrateForcedDischargeBulletproof.
      framework_substrate_pins_bulletproof_bundle]

info: PF/AxiomCheck_2026_06_23.lean:16:0:
```



```
'PF.Referee.CrossMillenniumInvariants_Extended_2026_06_19.
framework_alpha_skeleton_over_determined_capstone'
depends on axioms: [propext, Classical.choice, Quot.sound]
```

Build completed successfully (4373 jobs).

The four named substrate-tier citation axioms appearing above are: `Hardy1914_published_theorem_substrate` (Wiles-pattern citation of an external published-and-proven theorem); `Mayer1991_Cohen2025_substrate_HP_pr` (substrate-tier citation of the published-open Hilbert–Pólya program conjecture (C1)); and `framework_substrate_pins_bulletproof_bundle` (the substrate’s record-typed packaging of the three named open conjectures (C1), (C2), (C3) consumed by the six-axis closure theorem). The full corpus contains exactly four named project axioms; the fourth, `Mayer1991_Cohen2025_T3_sym_spectral`, appears in the substrate’s T_3^{sym} spectral-data variant of the RH route, not on the literal-mathlib-carrier chain above. A reader can verify there are no other axioms via `grep -rE "^axiom [a-zA-Z_][a-zA-Z0-9_]* :\" PF_Lean4_Code/PF/` which returns exactly four matches.

APPENDIX B. SEVENTEEN EXTENDED ALGEBRAIC INVARIANTS ON THE SUBSTRATE’S α -SKELETON

The paper’s §3 states that the substrate’s nine-tuple satisfies seventeen further algebraic invariants beyond the twelve baseline identities (I1)–(I12), bundled as `framework_alpha_skeleton_over_determined` in `PF/Referee/CrossMillenniumInvariants_Extended_2026_06_19.lean`. This appendix enumerates them. Combined with (I1)–(I12), the substrate’s α -skeleton simultaneously satisfies twenty-nine algebraic identities on nine real unknowns. Each (R1)–(S2) clause is a kernel-only theorem in `PF.CrossMillenniumMoreInvariants`; the bundle theorem in the file above re-exports them all as a single citable `Prop`.

Reciprocals (R1)–(R5):

$$\begin{aligned} \text{(R1)} \quad 1/\alpha_P &= \alpha_P/2 = \sqrt{2}/2, & \text{(R2)} \quad 1/\alpha_{\text{RH}} &= 2/3, \\ \text{(R3)} \quad 1/\alpha_{\text{YM}} &= 1/2, & \text{(R4)} \quad 1/\alpha_{\text{BSD}} &= 4/(3\pi), \\ \text{(R5)} \quad 1/\alpha_{\text{NS}} &= 2/(3\pi). \end{aligned}$$

Higher powers (P1)–(P6):

$$\begin{aligned} \text{(P1)} \quad \alpha_P^3 &= 2\alpha_P, & \text{(P2)} \quad \alpha_{\text{RH}}^3 &= 27/8, \\ \text{(P3)} \quad \alpha_{\text{YM}}^3 &= 8, & \text{(P4)} \quad \alpha_{\text{Hodge}}^3 &= 2\alpha_{\text{Hodge}} + 1, \\ \text{(P5)} \quad \alpha_{\text{Hodge}}^4 &= 3\alpha_{\text{Hodge}} + 2, & \text{(P6)} \quad \alpha_{\text{QG}}^4 &= 4\pi^2. \end{aligned}$$

Mixed products (M1)–(M4):

$$\begin{aligned} \text{(M1)} \quad \alpha_{\text{Hodge}} \cdot \alpha_{\text{NP}} &= (5/4) \alpha_{\text{Hodge}} + 1, & \text{(M2)} \quad \alpha_{\text{NP}}^2 &= (3/2) \alpha_{\text{Hodge}} + 17/16, \\ \text{(M3)} \quad \alpha_{\text{RH}} \cdot \alpha_{\text{BSD}} &= 9\pi/8, & \text{(M4)} \quad \alpha_{\text{YM}} \cdot \alpha_{\text{BSD}} &= \alpha_{\text{NS}}. \end{aligned}$$

Sums (S1)–(S2):

$$\begin{aligned} \text{(S1)} \quad \alpha_{\text{NS}} + \alpha_{\text{BSD}} &= 9\pi/4, & \text{(S2)} \quad 2\alpha_{\text{BSD}} &= \alpha_{\text{NS}}. \end{aligned}$$

The substrate’s α -skeleton is not a free-parameter assignment; it is the unique positive simultaneous solution of (I1)–(I12) over $\mathbb{R}^9$, and the same nine-tuple satisfies seventeen further algebraic identities (R1)–(S2) kernel-only proven on the constructed solution. `cross_millennium_invariants_extended` in `PF/Referee/CrossMillenniumInvariants_Extended_2026_06_19.lean` re-exports the entire (R1)–(S2) set as a single citable theorem.

The α_{NP} power-expansion recurrence. The substrate’s α_{NP} value sits in the rank-2 $\mathbb{Q}$ -module $\mathbb{Q} \cdot \alpha_{\text{Hodge}} \oplus \mathbb{Q} \cdot 1$; every power $\alpha_{\text{NP}}^k = a_k \cdot \alpha_{\text{Hodge}} + b_k$ with $a_k, b_k \in \mathbb{Q}$ satisfies the kernel-only recurrence

$$a_{k+1} = \frac{5}{4} a_k + b_k, \quad b_{k+1} = a_k + \frac{1}{4} b_k.$$

The 2×2 transition matrix $M = \begin{pmatrix} 5/4 & 1 \\ 1 & 1/4 \end{pmatrix}$ realises multiplication by α_{NP} on the rank-2 $\mathbb{Q}$ -module $\mathbb{Q} \cdot \alpha_{\text{Hodge}} \oplus \mathbb{Q} \cdot 1$; its characteristic polynomial is therefore the minimal polynomial of α_{NP} over

$\mathbb{Q}$: $\lambda^2 - \frac{3}{2}\lambda - \frac{11}{16} = 0$, equivalently $16\lambda^2 - 24\lambda - 11 = 0$ — *the substrate's NP polylog quadratic itself*. The eigenvalues are α_{NP} and its Galois conjugate $(3 - 2\sqrt{5})/4$. The substrate's NP algebraic signature is the same algebraic object viewed at the recurrence level: the minimal polynomial of α_{NP} , the characteristic polynomial of the power-expansion transition, and the substrate-derived NP polylog quadratic are one identity. Three new kernel-only theorems in `PF/AlphaNPHigherPowers_2026_06_25.lean`: `$\alpha_{\text{NP\_cubed}}$` giving $\alpha_{\text{NP}}^3 = (47/16)\alpha_{\text{Hodge}} + 113/64$; `$\alpha_{\text{NP\_fourth}}$` giving $\alpha_{\text{NP}}^4 = (87/16)\alpha_{\text{Hodge}} + 865/256$; `$\alpha_{\text{NP\_power\_recurrence}}$` giving the general recurrence step.

APPENDIX C. SQUARED-SPECTRUM, $\lambda_0(\text{NP})$ OVER $\mathbb{Q}(\sqrt{5}) \cdot \pi$, AND THE SUBSTRATE'S P/NP SPECTRAL-GAP CLOSED FORM

The substrate's universal coupling $\lambda_0(\alpha) = \pi/(10\alpha)$ on each class, composed with the algebraic forms of α from Table 1, yields explicit closed forms for the squared spectrum, for $\lambda_0(\text{NP})$ over $\mathbb{Q}(\sqrt{5}) \cdot \pi$, and for the P/NP spectral gap $\Delta = \lambda_0(\text{P}) - \lambda_0(\text{NP})$. The content of this appendix is all kernel-only in the Lean files `PF/Lambda0SquaredClosedForms_2026_06_25.lean`, `PF/Lambda0CrossClassStructure_2026_06_25.lean`, and `PF/SpectralGapClosedForm_2026_06_25.lean`, all imported by `PF.lean` at HEAD.

The substrate's squared spectrum partitions cleanly. The closed-form values of $\lambda_0(\alpha)^2$ across the nine substrate classes fall into three structural strata over the substrate's basis $\mathcal{B}$:

Stratum	Class	Closed form
$\mathbb{Q}$	NS	$1/225$
$\pi \cdot \mathbb{Q}$	QG (<i>unique</i>)	$\pi/200$
$\pi^2 \cdot \mathbb{Q}$	Poincaré, RH, P, YM	$\pi^2/100, \pi^2/225, \pi^2/200, \pi^2/400$
$\pi^2 \cdot \mathbb{Q}(\sqrt{5})$	Hodge	$\pi^2(3 - \sqrt{5})/200$

The quantum-gravity class is the only class whose squared eigenvalue scales as π^1 , not π^2 or rational, because $\alpha_{\text{QG}}^2 = 2\pi$ enters as a factor in the denominator of $\lambda_0(\text{QG})^2$. The corresponding kernel-only theorems are `lambda_0_Poincare_sq`, `lambda_0_RH_sq`, `lambda_0_P_sq`, `lambda_0_YM_sq`, `lambda_0_NS_sq`, `lambda_0_QG_sq`, `lambda_0_Hodge_sq`; they are bundled as `lambda_0_squared_spectrum_capstone`. Two structural relations follow at the kernel: $\pi \cdot \lambda_0(\text{QG})^2 = \lambda_0(\text{P})^2$ (theorem `lambda_0_P_sq_eq_pi_times_lambda_0_QG_sq`) and $\lambda_0(\text{YM})^2 = \lambda_0(\text{Poincaré})^2/4$. The single π -scaled relation between P and QG collapses two of the substrate's three irrational classes into one π -multiplicative tie.

$\lambda_0(\text{NP})$ over $\mathbb{Q}(\sqrt{5}) \cdot \pi$. Rationalising $\lambda_0(\text{NP}) = \pi/(10(\varphi + 1/4))$ against the Galois conjugate of $3 + 2\sqrt{5}$:

$$\lambda_0(\text{NP}) = \frac{\pi}{10(\varphi + 1/4)} = \frac{2\pi(2\sqrt{5} - 3)}{55}.$$

Kernel-only as `lambda_0_NP_closed_form_sqrt5`. The substrate's NP class lives explicitly in the field $\mathbb{Q}(\sqrt{5}) \cdot \pi$ at the λ_0 level. Cross-class structural identities on the rational pair (NS, BSD): $\lambda_0(\text{NS}) \cdot \lambda_0(\text{BSD}) = 2/225$ (no π), $\lambda_0(\text{NS}) + \lambda_0(\text{BSD}) = 1/5$ (no π). Mixed-stratum products: $\lambda_0(\text{RH}) \cdot \lambda_0(\text{BSD}) = 2\pi/225$ (kernel-only as `lambda_0_RH_mul_lambda_0_BSD`); $\lambda_0(\text{YM}) \cdot \lambda_0(\text{NS}) = \pi/300$. These cross-class products structurally land in $\mathbb{Q} \oplus \pi\mathbb{Q} \oplus \pi^2\mathbb{Q} \oplus \mathbb{Q}(\sqrt{5}) \cdot \pi$ — the substrate's λ_0 tensor structure over $\mathcal{B}$.

The substrate's P/NP spectral-gap closed form. The substrate's complexity-class separation $\alpha_{\text{P}} \neq \alpha_{\text{NP}}$ (§3, strict total ordering) lifts to an explicit closed form for the ground-state spectral gap:

$$\Delta := \lambda_0(\text{P}) - \lambda_0(\text{NP}) = \frac{\pi}{10\sqrt{2}} - \frac{\pi}{10(\varphi + 1/4)} = \frac{\pi(24 + 11\sqrt{2} - 16\sqrt{5})}{220}.$$

The substrate's spectral gap lives in $\mathbb{Q}(\sqrt{2}, \sqrt{5}) \cdot \pi$; the numerator $24 + 11\sqrt{2} - 16\sqrt{5}$ is an algebraic integer in $\mathbb{Z}[\sqrt{2}, \sqrt{5}]$. The closed form is the kernel-only theorem `spectral_gap_closed_form`, and the positivity of the numerator (and hence of Δ , given $\pi > 0$) is `spectral_gap_numerator_positive`: the positivity reduces to $(24 + 11\sqrt{2})^2 > (16\sqrt{5})^2$, i.e. $818 + 528\sqrt{2} > 1280$, i.e. $\sqrt{2} > 7/8$, which

is immediate from $\sqrt{2} > 1$. Bundle: `spectral_gap_closed_form_capstone`. The substrate’s P vs. NP separation is therefore not only strict; it is an explicit element of $\mathbb{Q}(\sqrt{2}, \sqrt{5}) \cdot \pi$ with a known closed form.

ACKNOWLEDGMENTS

To Liz, who carried this household while the substrate was being formalised. To Dylan, who would not let me stop. To Lily, who I hope reads this one day. To the open-source Lean 4 community and the mathlib4 maintainers, whose work made this verification possible.

REFERENCES

- [1] P. Cohen. *Principia Fractalis: Fractal Resonance Mathematics, Second Edition*. Version 2.6.1, 915 pages, 2026. <https://github.com/FractalDevTeam/Principia-Fractalis>
- [2] G. Perelman. The entropy formula for the Ricci flow and its geometric applications. *arXiv:math/0211159*, 2002; and subsequent works, 2003.
- [3] G. H. Hardy. Sur les zéros de la fonction $\zeta(s)$ de Riemann. *Comptes Rendus Acad. Sci. Paris* 158, 1012–1014, 1914.
- [4] D. H. Mayer. The thermodynamic formalism approach to Selberg’s zeta function for $\mathrm{PSL}(2, \mathbb{Z})$. *Bull. Amer. Math. Soc.* 25 (1) 55–60, 1991.
- [5] A. Connes. Trace formula in noncommutative geometry and the zeros of the Riemann zeta function. *Selecta Math.* 5, 29–106, 1999.
- [6] L. de Moura and S. Ullrich. The Lean 4 theorem prover and programming language. In *Automated Deduction — CADE 28*, Springer, 2021.
- [7] N. Bourbaki. *Groupes et algèbres de Lie, Chapitres 4–6*. Hermann, Paris, 1968. Coxeter number $h(H_3) = 10$ tabulated in Ch. VI, §4.10.
- [8] L. Babai. Graph isomorphism in quasipolynomial time. In *Proc. 48th ACM Symp. Theory of Computing (STOC)*, 684–697, 2016. [arXiv:1512.03547](https://arxiv.org/abs/1512.03547).
- [9] U. Schöning. Graph isomorphism is in the low hierarchy. *J. Comput. Syst. Sci.* 37 (3) 312–323, 1988.
- [10] S. Goldwasser and M. Sipser. Private coins versus public coins in interactive proof systems. In *Proc. 18th ACM Symp. Theory of Computing (STOC)*, 59–68, 1986.
- [11] M. V. Berry and J. P. Keating. The Riemann zeros and eigenvalue asymptotics. *SIAM Review* 41 (2) 236–266, 1999.
- [12] J.-B. Bost and A. Connes. Hecke algebras, type III factors and phase transitions with spontaneous symmetry breaking in number theory. *Selecta Math. (N.S.)* 1 (3) 411–457, 1995.
- [13] A. Wiles. Modular elliptic curves and Fermat’s Last Theorem. *Annals of Math.* 141 (3) 443–551, 1995.
- [14] C. Fefferman. Existence and smoothness of the Navier–Stokes equation. Clay Mathematics Institute Millennium Problem statement, 2000. <https://www.claymath.org/wp-content/uploads/2022/06/navierstokes.pdf>
- [15] H. Fujita and T. Kato. On the Navier–Stokes initial value problem I. *Arch. Rational Mech. Anal.* 16, 269–315, 1964.
- [16] A. Jaffe and E. Witten. Quantum Yang–Mills theory. Clay Mathematics Institute Millennium Problem statement, 2006. <https://www.claymath.org/wp-content/uploads/2022/06/yangmills.pdf>
- [17] B. J. Birch and H. P. F. Swinnerton-Dyer. Notes on elliptic curves II. *J. Reine Angew. Math.* 218, 79–108, 1965.
- [18] W. V. D. Hodge. The topological invariants of algebraic varieties. In *Proc. Int. Congress of Mathematicians 1950*, vol. 1, 182–192, 1952.
- [19] C. Voisin. Some aspects of the Hodge conjecture. *Japan. J. Math.* 2 (2) 261–296, 2007.
- [20] S. Lefschetz. *L’analyse situs et la géométrie algébrique*. Gauthier-Villars, Paris, 1924.
- [21] The LMFDB Collaboration. *The L-functions and Modular Forms Database*. 2024. <https://www.lmfdb.org>
- [22] I. Esteban, M. C. Gonzalez-Garcia, M. Maltoni, T. Schwetz, A. Zhou. The fate of hints: updated global analysis of three-flavor neutrino oscillations (NuFit 6.0). *J. High Energy Phys.* 09, 178, 2024. <https://www.nu-fit.org/>
- [23] DESI Collaboration: A. G. Adame et al. DESI 2024 VI: cosmological constraints from the measurements of baryon acoustic oscillations. *J. Cosmol. Astropart. Phys.* 02, 021, 2025. [arXiv:2404.03002](https://arxiv.org/abs/2404.03002).
- [24] D. Scolnic et al. The Pantheon+ analysis: the full data set and light-curve release. *Astrophys. J.* 938 (2) 113, 2022. [arXiv:2112.03863](https://arxiv.org/abs/2112.03863).
- [25] S. More et al. (HSC Collaboration). Hyper Suprime-Cam Year 3 results: measurements of clustering of SDSS-BOSS galaxies, galaxy–galaxy lensing, and cosmic shear. *Phys. Rev. D* 108, 123520, 2023. [arXiv:2304.00702](https://arxiv.org/abs/2304.00702).
- [26] F. Spite and M. Spite. Abundance of lithium in unevolved halo stars and old disk stars: interpretation and consequences. *Astron. Astrophys.* 115, 357–366, 1982.
- [27] B. D. Fields, K. A. Olive, T.-H. Yeh, C. Young. Big-bang nucleosynthesis after Planck. *J. Cosmol. Astropart. Phys.* 03, 010, 2020. [arXiv:1912.01132](https://arxiv.org/abs/1912.01132).

- [28] LIGO Scientific, Virgo, and KAGRA Collaborations: R. Abbott et al. GWTC-3: compact binary coalescences observed by LIGO and Virgo during the second part of the third observing run. *Phys. Rev. X* 13, 041039, 2023. arXiv:2111.03606.
- [29] LIGO Scientific, Virgo, and KAGRA Collaborations: R. Abbott et al. GWTC-4.0: compact binary coalescences observed by LIGO and Virgo during the first part of the fourth observing run. arXiv:2508.18083, 2025.
- [30] The Mathlib Community. The Lean Mathematical Library. In *Proc. 9th ACM SIGPLAN Int. Conf. on Certified Programs and Proofs (CPP)*, 367–381, 2020.

MESA, ARIZONA, USA

Email address: `psolorzano@gmail.com`

URL: ORCID: ~0009-0002-0734-5565